3.4 Function Notation

Algebra I 3.4 · Student sheet

Name _____ Period ____ Date _____

I can…

- evaluate functions using function notation.
- find the input that gives a certain output.
- graph linear functions and interpret function notation in real life.

Part 1 · Function machine

$f(x) = 2x + 5$ $g(x) = x^2 - 1$

Input x	f(x)	g(x)

Output f(x)	Input x
15	
−3	

Part 2 · Notes: fill in the blanks

f(x) is read "_____."

To evaluate f(3), _____ 3 for x.

To find x when f(x) = 15, set the rule equal to 15 and _____.

f(3) = 11 means the point _____ is on the graph.

Part 3 · Examples (work them with the class)

Example 1

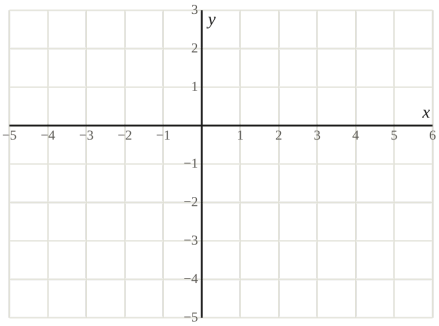
Let $f(x) = -4x + 7$. Find f(2) and f(−3).

Example 2

Let $g(x) = 5x - 3$. Find x so that g(x) = 22.

Example 3

Graph $f(x) = \frac{1}{2}x - 2$. Use the graph to find $f(4)$ and the x-value where $f(x) = -3$.



Example 4

The cost (in dollars) of renting a truck and driving m miles is $C(m) = 0.25m + 15$. Find $C(100)$ and explain what it means.

Part 4 · Partner practice

Partner 1

Let $h(x) = x^2 + 1$. Find $h(-3)$.

Partner 2

Let $f(x) = 3x + 4$. Find x so that $f(x) = -8$.